

Causal Inference & ML

TA Session 1

Data Science for Decision Making

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TA Materials:

All the material for TA Sessions — slides, and code will be hosted in the [GitHub repository](#).

```
1 git clone https://github.com/luciasauer/bse_causal_inference_ml.git
```

The TA sessions will mainly focus on **Stata**, but some examples in **Python** will also be provided.



Ensure that the required Python packages are installed:

Install uv

```
1 uv sync
```

Pull the repository and get the updated materials for each session.

Outline

- Panel Data
- Why Panel Data?
- Static and Linear Models
 - Pooled OLS
 - Fixed Effects Models
 - Random Effects Models
 - Hausman Test
- Dynamic Panel Data Models
 - Arellano-Bond Estimator

What panel data looks like...

- Panel data consists of observations on multiple entities over multiple time periods.
- Each entity is observed at several time points, allowing us to analyze changes over time within the same entities.

Entity ID	Time Period	Income	gender
1	2000	30	male
1	2001	33	male
1	2002	33	male
2	2000	50	female
2	2001	58	female
2	2002	61	female
3	2000	25	female
3	2001	30	female
3	2002	42	female

Wide Format Panel Data

- In practice, datasets such as those from the World Bank are typically organized in **wide format**.
- To perform panel data analysis, we first reshape the data into **long format**, so that each row corresponds to a specific entity and time period.

	Country	2000	2001	2002	2003	2004	2005
0	A	4058.14	4093.61	4821.78	4430.52	4981.48	3668.82
1	B	23692.66	24545.37	25319.78	24818.70	25805.80	27093.83
2	C	15288.34	14714.42	15167.32	15475.99	15806.36	17222.30
3	D	22256.36	23299.13	23038.89	23134.99	24062.82	25342.87
4	E	29145.24	30223.65	31543.82	31838.64	33580.40	34576.21
5	F	17686.25	17625.59	18690.89	18853.40	18500.80	20593.65

Long Format Panel Data

Stata use the command `reshape` to convert wide format data to long format:

```
1 gen id = _n
2 order id
3 reshape long x , i(id) j(year)
```

In Python we can use the `melt` function in pandas:

```
1 long = (
2     wide.melt("Country", var_name="year", value_name="gdp")
3     .assign(year=lambda d: d.year.astype(int))
4 )
5 long = long.sort_values(["Country", "year"])
```

	Country	year	gdp
0	A	2000	4058.14
7	A	2001	4093.61
14	A	2002	4821.78
21	A	2003	4430.52

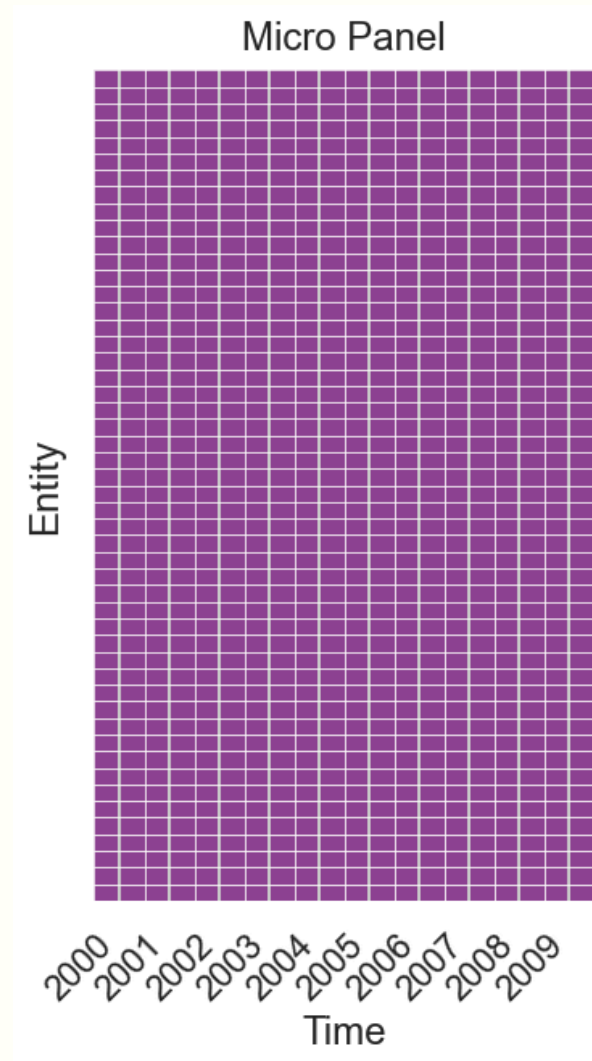
Types of Panel Data



Usefull package in R to plot panel data structure: [panelview](#)

Types of Panel Data

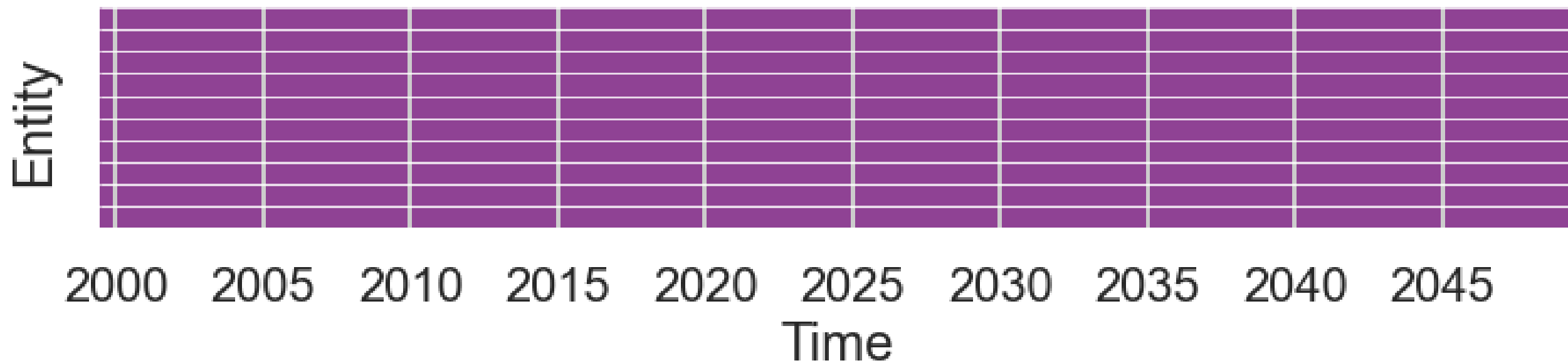
Micro panels: Large $N = 50$, Small $T = 10$



Types of Panel Data

Macro panels: small $N = 10$ and large $T = 50$

Macro Panel



Why Panel Data?

The problem

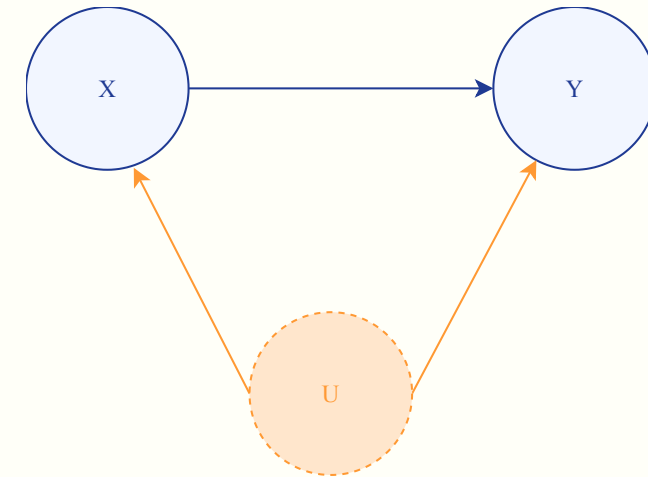
OLS requires exogeneity: $E(u_{it} | X_{it}) = 0$

When unobserved factors (c_i) affect (y_{it}) and are correlated with (X_{it}),

$$\text{Cov}(X_{it}, c_i) \neq 0$$

OLS suffers from **Omitted Variable Bias**.

- **With a single cross section**, some remedies are Instrumental Variables (IV)
- **With panel data**, we can control for **time-invariant unobserved heterogeneity**



Panel Data Models

Throughout this session, we focus on **panel data models** designed to handle unobserved heterogeneity and dynamic behavior over time.

1. Static Linear Panel Models

- **Pooled OLS** — Ignores unobserved heterogeneity
- **Fixed Effects (FE)** — Allows $\text{Cov}(c_i, X_{it}) \neq 0$
- **Random Effects (RE)** — Requires $\text{Cov}(c_i, X_{it}) = 0$

2. Dynamic Panel Models

- Arellano–Bond (Difference GMM)
- System GMM

Used when the model includes **lagged dependent variables** and standard FE is biased.

1. Static and Linear Models

The Grunfeld Investment Data

We use the classic **Grunfeld (1958)** dataset, which contains annual data on 10 large US manufacturing firms from 1935 to 1954.

To access the data in Stata, use:

```
1 webuse grunfeld, clear
```

In Python:

```
1 grunfeld = pd.read_stata("data/grunfeld.dta")
2 grunfeld.head(5)
```

	company	year	invest	mvalue	kstock	time
0	1	1935-01-01	317.600006	3078.500000	2.800000	1
1	1	1936-01-01	391.799988	4661.700195	52.599998	2
2	1	1937-01-01	410.600006	5387.100098	156.899994	3
3	1	1938-01-01	257.700012	2792.199951	209.199997	4
4	1	1939-01-01	330.799988	4313.200195	203.399994	5

The model

We want to model the investment behavior of firms over time:

$$invest_{it} = \beta_0 + \beta_1 mvalue_{i,t-1} + \beta_2 kstock_{i,t-1} + c_i + \varepsilon_{it}$$

- $invest_{it}$: Investment (current year)
- $mvalue_{i,t-1}$: Market value of the firm (end of previous year)
- $kstock_{i,t-1}$: Capital stock (end of previous year)
- c_i : Unobserved time-invariant firm-specific heterogeneity (e.g., management quality, corporate culture, industry position).

Research question

How do market value and capital stock affect firm investment?

Setting Up the Panel Structure

In Stata, we declare the panel structure using the `xtset` command:

```
1 xtset company year
```

```
Panel variable: company (strongly balanced)
Time variable: year, 1935 to 1954
Delta: 1 year
```

Interpretation: we have a *strongly balanced* panel — all 10 firms are observed in all 20 years, giving us $N = 10 \times 20 = 200$ observations.

In Python, we can set the index to be a MultiIndex with company and year:

```
1 grunfeld = grunfeld.set_index(['company', 'year'])
```

Descriptive Statistics

In Stata we can explore pooled summary statistics with the `summarize` command:

```
1 summarize invest mvalue kstock
```

Variable	Obs	Mean	Std. dev.	Min	Max
-----+-----					
invest	200	145.9583	216.8753	.93	1486.7
mvalue	200	1081.681	1314.47	58.12	6241.7
kstock	200	276.0172	301.1039	.8	2226.3

In Python, we can compute the same summary statistics using the `describe` method:

```
1 grunfeld.reset_index()[['invest', 'mvalue', 'kstock']].describe()
```

	invest	mvalue	kstock
count	200.000000	200.000000	200.000000
mean	145.958237	1081.681152	276.017151
std	216.875290	1314.469727	301.103912
min	0.930000	58.119999	0.800000
25%	33.557499	199.975006	79.174999
50%	57.485001	517.950012	205.599998
75%	138.040001	1679.850006	358.099991
max	1486.699951	6241.700195	2226.300049

Sources of variation in Panel Data

A key feature of panel data is that variation comes from 2 sources:

between = $\bar{Y}_i - \bar{Y}$ and within = $Y_{it} - \bar{Y}_i$

Simplified example with only 3 firms over 3 years:

company	year	$invest_{it}$	$in\bar{vest}_i$	$invest_{it} - in\bar{vest}$	$invest_{it} - in\bar{vest}_i$	$in\bar{vest}_i - in\bar{vest}$
1	1935	317.6	373.33	60.89	−55.73	116.62
1	1936	391.8	373.33	135.09	18.47	116.62
1	1937	410.6	373.33	153.89	37.27	116.62
2	1935	209.9	345.03	−46.81	−135.13	88.32
2	1936	355.3	345.03	98.59	10.27	88.32
2	1937	469.9	345.03	213.19	124.87	88.32
3	1935	33.1	51.77	−223.61	−18.67	−204.94
3	1936	45.0	51.77	−211.71	−6.77	−204.94
3	1937	77.2	51.77	−179.51	25.43	−204.94

Within and Between Variation

This can be summarized using the `xtsum` command in Stata:

```
1 xtsum invest mvalue kstock
```

Variable		Mean	Std. dev.	Min	Max	Observations	
invest		145.9583	216.8753	.93	1486.7	N =	200
	between		198.8242	3.0845	608.02	n =	10
	within		106.1986	-204.3617	1024.638	T =	20
mvalue		1081.681	1314.47	58.12	6241.7	N =	200
	between		1334.917	70.921	4333.845	n =	10
	within		340.5421	-459.964	2989.536	T =	20
kstock		276.0172	301.1039	.8	2226.3	N =	200
	between		200.9701	5.9415	648.435	n =	10
	within		232.6603	-369.6179	1853.882	T =	20

Interpretation:

- For *invest*: between-firm SD (198) > within-firm SD (106). Most variation is across firms, not within firms over time.
- This suggests that firm-specific characteristics (captured by c_i) are important.
- Fixed effects will exploit only the within variation; random effects uses both.

First Case

- Pooled OLS ignores unobserved heterogeneity and pools all observations together.
- It assumes that $c_i = c$ is constant
- If c_i is correlated with *mvalue* or *kstock*, estimates are **biased**.

In Stata, we can estimate the pooled OLS model using the `reg` command with robust standard errors because of potential heteroskedasticity:

```
1 reg invest mvalue kstock, robust
```

```
Linear regression               Number of obs   =          200
                               F(2, 197)        =        178.83
                               Prob > F          =         0.0000
                               R-squared         =         0.8124
                               Root MSE      =         94.408
```

```
-----+-----
               |               Robust
invest | Coefficient  std. err.   t    P>|t|    [95% conf. interval]
-----+-----
mvalue |   .1155622   .006811   16.97  0.000   .1021304   .1289939
kstock |   .2306785   .0488655    4.72  0.000   .1343118   .3270452
 _cons |  -42.71437   11.5747   -3.69  0.000  -65.54059  -19.88815
-----+-----
```

Estimation Methods - Fixed Effects (Within Estimator)

1. We start from the panel data model:

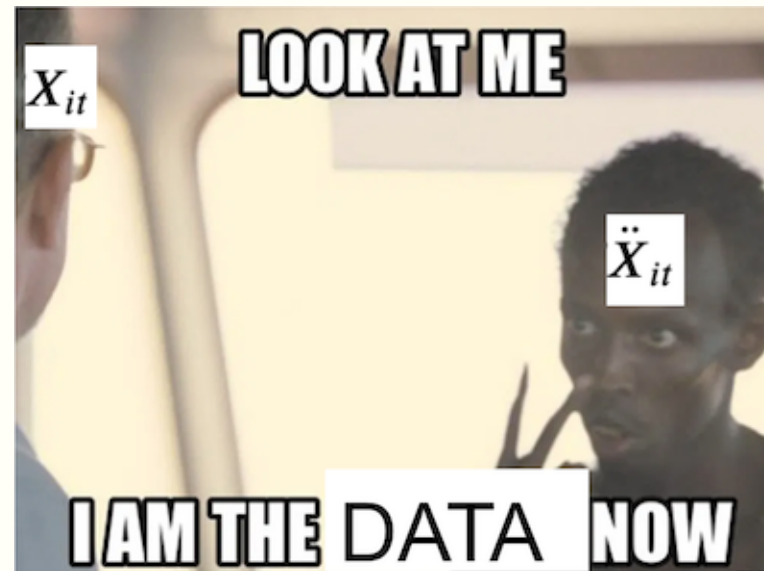
$$invest_{it} = c_i + \beta_1 mvalue_{i,t-1} + \beta_2 kstock_{i,t-1} + \varepsilon_{it}$$

where we allow for c_i to be correlated with $mvalue$ and $kstock$, random and unobserved.

2. Fixed Effects removes c_i by comparing each unit **to itself over time**.

$$invest_{it} - \bar{invest}_i = \beta_1 (mvalue_{i,t-1} - \bar{mvalue}_i) + \beta_2 (kstock_{i,t-1} - \bar{kstock}_i) + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

3. Apply OLS to the demeaned variables.



Estimation Methods - Fixed Effects (Within Estimator)

All these steps are done automatically in Stata with the `xtreg, fe` command.

```
1 xtreg invest mvalue kstock, fe robust
```

Also if we want to estimate FE by Dummy Variable approach:

```
1 reg invest mvalue kstock i.company, vce(cluster company)
```

To implement FE - Within Estimator in Python, we can use the `linearmodels.panel` package:

```
1 from linearmodels.panel import PanelOLS
2 fe_model = PanelOLS.from_formula(
3     'invest ~ 1 + mvalue + kstock + EntityEffects',
4     data=grunfeld
5 )
6 fe_results = fe_model.fit(cov_type='clustered', cluster_entity=True)
7 print(fe_results)
```

Interpretation of Fixed Effects Results

```
Fixed-effects (within) regression      Number of obs   =       200
Group variable: company               Number of groups =       10

R-squared:                            Obs per group:
    Within = 0.7668                    min =       20
    Between = 0.8194                  avg =      20.0
    Overall = 0.8060                  max =       20

corr(u_i, Xb) = -0.1517                F(2, 9)          =      28.31
                                      Prob > F           =      0.0001
```

(Std. err. adjusted for 10 clusters in company)

```
-----
            |               Robust
            | Coefficient std. err.      t    P>|t|    [95% conf. interval]
-----+-----
    mvalue |   .1101238   .0151945    7.25  0.000    .0757515   .1444961
    kstock |   .3100653   .0527518    5.88  0.000    .1907325   .4293981
      _cons |  -58.74393  27.60286   -2.13  0.062   -121.1859   3.698079
-----+-----
    sigma_u |  85.732501
    sigma_e |  52.767964
      rho   |   .72525012  (fraction of variance due to u_i)
-----
```

- *mvalue*: A \$1 million increase in market value is associated with a \$0.11 million increase in investment.
- *kstock*: A \$1 million increase in capital stock is associated with a \$0.31 million increase in investment.
- $\rho = 0.73$: 73% of the total variance is due to differences *between* firms (unobserved heterogeneity).
- $\text{corr}(u_i, Xb) = 0.46$: The firm effects are correlated with the regressors, justifying the use of FE over RE.

Estimation Methods - Random Effects Model

```
1 xtreg invest mvalue kstock, re robust
```

```
Random-effects GLS regression              Number of obs   =          200
Group variable: company                   Number of groups  =           10
R-squared:                               Obs per group:
    Within   = 0.7668                      min =          20
    Between  = 0.8196                      avg  =         20.0
    Overall  = 0.8061                      max  =          20
                                           Wald chi2(2)      =         70.13
corr(u_i, X) = 0 (assumed)                Prob > chi2       =         0.0000
```

(Std. err. adjusted for 10 clusters in company)

		Robust				
invest	Coefficient	std. err.	z	P> z	[95% conf. interval]	

mvalue	.1097811	.0137557	7.98	0.000	.0828206	.1367417
kstock	.308113	.0549728	5.60	0.000	.2003683	.4158576
_cons	-57.83441	24.84323	-2.33	0.020	-106.5262	-9.142576

sigma_u	84.20095					
sigma_e	52.767964					
rho	.71800838	(fraction of variance due to u_i)				

- RE assumes $\text{Cov}(c_i, X_{it}) = 0$.
- The Hausman test will help us choose between FE and RE.

Fixed Effects vs. Random Effects: The Hausman Test

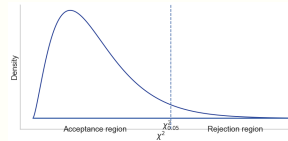
The Hausman test compares the FE and RE estimators.

$$\begin{aligned} H_0 : \text{Cov}(c_i, X_{it}) &= 0 && \text{(both estimators are consistent, RE is more efficient)} \\ H_A : \text{Cov}(c_i, X_{it}) &\neq 0 && \text{(only FE is consistent)} \end{aligned}$$

Statistic:

$$H = (\hat{\beta}_{RE} - \hat{\beta}_{FE})' [\text{Var}(\hat{\beta}_{FE}) - \text{Var}(\hat{\beta}_{RE})]^{-1} (\hat{\beta}_{RE} - \hat{\beta}_{FE})$$

This statistic follows a chi-squared distribution under the null hypothesis and RE1 - RE3.



- If H is large, then there is a statistical difference between these two estimators, and we reject H_0 in favor of FE.
- If H is small, we fail to reject H_0 and prefer RE for efficiency.

The Hausman Test in Stata

In Stata, we can perform the Hausman test with `hausman` command

1. Estimate and store the FE model

```
1 quietly xtreg invest mvalue kstock, fe
2 estimates store FE
```

2. Estimate and store the RE model

```
1 quietly xtreg invest mvalue kstock, re
2 estimates store RE
```

3. Run the Hausman test

```
1 * Hausman test
2 hausman FE RE, sigmamore
```

- However, this test assumes homoskedasticity and no clustering. If these assumptions are violated, the standard Hausman test may be invalid.
- In such cases, consider using robust or clustered versions of the Hausman test, with `xtoverid` command.
- Nevertheless, FE is often preferred in practice due to is consistent whether or not $\text{Cov}(c_i, X_{it}) = 0$.

Clustered Standard Errors: Why and How

In panel data, observations within the same entity (firm, country, individual) are typically **not independent**. This violates a key OLS assumption and causes problems:

- **Heteroskedasticity:** Error variance may differ across entities
- **Serial correlation:** Errors for the same entity are correlated over time

For pooled OLS:

```
1 *Heteroskedastic robust SE
2 reg invest mvalue kstock, robust
3
4 * Cluster-robust: handles BOTH heteroskedasticity AND serial correlation
5 reg invest mvalue kstock, vce(cluster company)
```

For FE and RE models:

```
1 *Heteroskedastic robust SE
2 xtreg invest mvalue kstock, fe robust
3
4 * Cluster-robust: handles BOTH heteroskedasticity AND serial correlation
5 xtreg invest mvalue kstock, fe vce(cluster company)
```

Point estimates remain the same; only standard errors and inference change.

How many clusters are enough?

Cluster-robust standard errors rely on asymptotic theory as the number of clusters $G \rightarrow \infty$. With few clusters:

- $G \geq 50 \rightarrow$ generally safe
- $20 \leq G < 50 \rightarrow$ use with caution
- $G < 20 \rightarrow$ clustered SEs may be unreliable

👉 *Grunfeld data*: $G = 10$ **firms** $\rightarrow$ inference is fragile

With few clusters: use wild cluster bootstrap

```
1 * Wild cluster bootstrap (requires boottest package)
2 ssc install boottest
3 xtreg invest mvalue kstock, fe
4 boottest mvalue, cluster(company) reps(999)
```

Multiple clustering dimensions:

The `reghdfe` command allows for multiple fixed effects and multi-way clustered standard errors in a computationally efficient way.

```
1 reghdfe invest mvalue kstock, absorb(company) vce(cluster company year)
```

Adding Time Fixed Effects: Two-Way Fixed Effects

There may be common shocks affecting all entities in a given time period (e.g., economic crises, policy changes). To control for these, we can include time fixed effects:

```
1 xtreg invest mvalue kstock i.year, fe robust
```

Testing for Time Fixed Effects

```
1 testparm i.year
```

```
F( 9, 9) = 6.15  
Prob > F = 0.0062
```

Interpretation: The null hypothesis is significant at the 0.6%, meaning that year coefficients are jointly statistically significant. Time fixed effects are necessary for this dataset.

2. Dynamic Models

Introduction to Dynamic Panel Data Models

In this section, we'll understand why lagged dependent variables break our usual toolkit and how GMM methods solve this problem.

Following Arellano and Bond (1991), we will estimate a dynamic labor demand model using the `abdata` dataset of UK firms from 1976 to 1984.

The model:

$$n_{it} = c_i + \rho \cdot n_{it-1} + X'_{it} \cdot \beta + \epsilon_{it}$$

where:

- n_{it} : log employment
- ρ : **persistence parameter** (key!)
- c_i : firm fixed effect
- X_{it} : wages, capital, output in log (current and lagged)

Ignoring Dynamic Panel Structure

Approach 1: Pooled OLS

Ignore panel structure, estimate:

$$n_{it} = c_i + \rho \cdot n_{it-1} + X'_{it} \cdot \beta + u_{it}$$

```
1 reg n nL1 nL2 w wL1 k kL1 kL2 ys ysL1 ysL2 yr*, cluster(id)
```

Problem: The composite error $u_{it} = c_i + \epsilon_{it}$ contains c_i (productivity, management quality...)

- If c_i is correlated with $n_{it-1} \rightarrow$ **OVB**
- Typically: more productive firms $\rightarrow$ hire more $\rightarrow Cov(n_{it-1}, c_i) > 0$
- Result: $\hat{\rho}_{OLS}$ is **biased upward** 

$\hat{\rho} = 1.044 (>1 !)$ — Not plausible for a stationary series!

Approach 2: Random Effects

Assume $\text{cov}(c_i, X_{it}) = 0$:

```
1 xtreg n nL1 nL2 w wL1 k kL1 kL2 ys ysL1 ysL2 yr*, re
```

Problem: Same as OLS!

Even if c_i were truly random (questionable!):

- n_{it-1} is mechanically correlated with c_i
- Because $n_{it-1} = c_i + \rho n_{it-2} + \dots$
- RE estimator is **inconsistent**

$\hat{\rho} = 1.044$ — Same as pooled OLS!

Approach 3: Fixed Effects (Within)

Demmean to eliminate c_i :

$$n_{it} - \bar{n}_i = \rho(n_{it-1} - \bar{n}_i) + (X_{it} - \bar{X}_i)' \beta + (\epsilon_{it} - \bar{\epsilon}_i)$$

```
1 xtreg n nL1 nL2 w wL1 k kL1 kL2 ys ysL1 ysL2 yr*, fe cluster(id)
```

Problem: Nickell Bias

After demeaning:

- $(n_{it-1} - \bar{n}_i)$ is correlated with $(\epsilon_{it} - \bar{\epsilon}_i)$
- Why? Because $\bar{n}_i = \frac{1}{T} \sum_{s=1}^T n_{is}$ includes n_{it} , which contains ϵ_{it}
- Result: $\hat{\rho}_{FE} = 0.732$ is **biased downward** ⬇️

$$\text{Nickell Bias} \approx -\frac{1 + \rho}{T - 1} \approx -\frac{1 + 0.8}{9 - 1} \approx -0.225$$

Solution: Arellano-Bond and System GMM

To address the **Nickell bias**, Arellano and Bond proposed the Arellano-Bond estimator, which uses a Generalized Method of Moments (GMM) approach.

1. **First Difference** to eliminate c_i :

$$\Delta n_{it} = \rho \Delta n_{it-1} + \Delta X'_{it} \beta + \Delta \epsilon_{it}$$

2. **Problem remains:** Δn_{it-1} is correlated with $\Delta \epsilon_{it}$

- Because $\Delta n_{it-1} = n_{it-1} - n_{it-2}$ contains ϵ_{it-1}
- And $\Delta \epsilon_{it} = \epsilon_{it} - \epsilon_{it-1}$ also contains ϵ_{it-1}

3. **Solution:** Use **lags in levels** as instruments!

$$E[n_{it-s} \cdot \Delta \epsilon_{it}] = 0 \quad \text{for } s \geq 2$$

- **Relevance:** n_{it-2} is correlated with Δn_{it-1} ✓
- **Exogeneity:** n_{it-2} is **not** correlated with $\Delta \epsilon_{it} = \epsilon_{it} - \epsilon_{it-1}$

Stata Implementation

To implement Arellano-Bond in Stata, we use the `xtabond2`

```
1 // Arellano-Bond estimation
2 xtabond2 n nL1 nL2 w wL1 k kL1 kL2 ys ysL1 ysL2 yr*, ///
3     gmm(L.(n w k), lag(2 .)) ///      // Treat as endogenous, lags 2+
4     iv(L(0/2).ys yr*) ///              // Treat as exogenous
5     noleveleq ///                      // Difference GMM only
6     robust ///                        // Robust standard errors
7     small                             // Small sample correction
```

Key Options

- `gmm(L.(n w k), lag(2 .))` : Use lags 2, 3, ... as instruments
- `iv(L(0/2).ys yr*)` : Exogenous variables (standard instruments)
- `noleveleq` : Exclude level equation (pure Difference GMM)

Results

Comparing the estimates:

Variable	Pooled OLS	Random Effects	Fixed Effects	Arellano–Bond
nL1	1.044***	1.044***	0.732***	0.817***

1. $\hat{\rho}_{AB} = 0.817$ lies between FE (0.732) and OLS (1.044) ✓

$$\hat{\rho}_{fe} < \hat{\rho}_{ab} < \hat{\rho}_{pooled,re}$$

- 2. Implies **strong employment persistence** (elasticity = 0.82)
- 3. AR(2) test: $p = 0.307 \rightarrow$ No second-order correlation ✓
- 4. Hansen test: $p = 0.174 \rightarrow$ Instruments are valid ✓

Why AB is in the middle:

- **Removes upward bias:** First differences eliminate c_i
- **Removes downward bias:** Instruments break the mechanical correlation